Robust Self-Paced Incremental Learning for Multitemporal Remote Sensing Image Classification

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***Abstract*— Classification of multitemporal remote sensing (MTRS) images with only a few labels of one of these images has attracted widespread interest in recent years. It usually confronts three problems: domain increment, class increment, and class disappearance. In this article, a robust self-paced incremental learning (RSPIL) is proposed to alleviate the above problems. First, change detection is used to transfer labels from the source image to the target one for formulating a combined training set. Then, the weighted classification loss and the distillation loss are considered to ensure classification performance and minimal forgetting. In particular, a novel entropy-inhibit loss is proposed to suppress the classification capability for the disappearing classes. These losses are combined with self-paced learning (SPL) by introducing a weight variable to measure the "easiness" of the training samples, which is able to automatically acquire accurate decision boundaries from easy to hard since the combined training set generated by change detection contains noisy samples and outliers. Finally, a nearest-average-eigenvector classifier and an exemplar set management strategy based on the sample weights are designed to alleviate catastrophic forgetting (CF). Twenty-four MTRS image datasets from four areas are considered in the experiments. The classification results demonstrate that the proposed method is able to alleviate CF and achieves significant improvements on multitemporal image datasets.**

***Index Terms*— Incremental learning, knowledge distillation, multitemporal images, self-paced learning (SPL).**

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I. Introduction

THE analysis of land cover maps derived from multitemporal remote sensing (MTRS) images can allow for the timely monitoring of land cover dynamics at a wide spatial scale, which can be used to assess damaged areas [1], analyze urban expansion [2], and investigate the environment [3]. The classification maps of the MTRS images can be obtained by supervised methods which require sufficient manual annotation information (label information) [4]. Unfortunately, the time and human labor costs for the labeling process are often unacceptable. As a result, it cannot be guaranteed to process sufficient training data for every image [5]. The temporal correlation and spectral similarity between MTRS images can be utilized to tackle the above issues [6]. In particular, after training on the source image (image with labels), classifiers are able to obtain the classification map of the target image (image without labels) [7]. However, due to different atmospheric conditions [8], seasonal variations affecting vegetation phenology, and other factors, even the data in MTRS images may follow different probability distributions, which is termed as cross-image dataset shifts [9], [10] and leads to the unsatisfactory classification results on the target image of the classifiers. Therefore, suitable domain adaptation (DA) strategies are needed to solve the dataset shifts [10], [11], [12], [13].

DA strategies aim to leverage label information from the source domain (source images) for the classification of the target domain (target images). Depending on the presence of labeled samples in the source and target domains [11], DA methods are generally categorized into three types: supervised DA, unsupervised DA, and semi-supervised DA. Unsupervised DA assumes that there are no labeled samples in either the source or target domains. By employing content-based methods and adversarial frameworks, Yu et al. [14] proposed an innovative unsupervised DA method that achieves feature alignment and significantly improves the classification performance of hyperspectral images. For synthetic aperture radar (SAR) images lacking labels, [15] enhanced the efficacy of SAR image retrieval by integrating domain discrimination and pseudo-label prediction with adversarial networks, and by leveraging category information obtained through clustering of SAR images. The fundamental assumption of semi-supervised

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DA is that the source domain possesses ample labeled data, while the target domain lacks these labels. Based on this assumption, semi-supervised DA strategies involve adapting the parameters of a classifier already trained on the source domain to fit the target domain [16], [17], [18]. This strategy [17] has been further expanded through the improvement of cascading classification Bayesian rules that leverage the temporal correlation between domains. The successful implementation of semi-supervised DA relies on two key assumptions [19]: 1) the source and target domains should contain the same set of categories and 2) the data distributions of these two domains should be sufficiently correlated.

However, in the analysis of MTRS images, the issue of interdomain class increment and disappearance is commonly encountered. Specifically, if a category exists in the source domain images but is absent in the target domain images, then that category is considered as disappeared. Conversely, if a category appears only in the target domain, it is deemed as an increment. Furthermore, the differences in data distributions between domains may be significant, making it challenging to satisfy the aforementioned assumptions. It has been demonstrated that the change-detection-driven transfer learning (CDTL) [20], [21] methods can overcome the aforementioned limitations. In the CDTL methods, the label information of the source domain image corresponding to the unchanged pixels determined by CD is transferred to the target domain image to formulate the training set.

DA and CDTL-based methods mentioned above effectively alleviate the issue of cross-image dataset shift, enabling models to adapt to the data distribution of the target domain and improving classification performance. However, these methods introduce a new challenge: when the model parameters are updated, the parameters tend to adapt to the data distribution of the target domain, potentially losing its adaptability to the source domain's data distribution. The performance of the source domain may decrease, which is termed as catastrophic forgetting (CF) and is particularly prominent in MTRS image classification [22]. Note that the new images in MTRS images may follow the data distribution which is related to those that have appeared. For example, seasonal variations, sensor parameters, and other factors can lead to periodic changes in the features of land cover categories, as well as the periodic appearance and disappearance of some categories. It is a natural idea to employ incremental learning approaches to alleviate CF. In general, these methods can be summarized in three forms [23], [24]: replay-based methods [25], [26], parameter isolation-based methods [27], [28], and regularization-based methods [22], [29].

However, there are several challenges in CDTL-based MTRS image classification algorithms that make incremental learning methods inadaptable. The problems of domain increment, class increment, and interdomain class disappearance occur in image sequences. Because of the domain increment, data represent the same category belonging to different domains. The decision boundaries on the source and target domains of the same class may change in the process of model training. It is a natural idea that the class incremental learning (CIL) methods are able to deal with the class increment problem in MTRS images. However, the training set of CIL methods contains data from both domains, and the dataset shifts may prevent the model from learning accurate decision boundaries. By automatically obtaining the "easiness" degree of training samples (sample weights), self-paced learning (SPL) [30] is able to dynamically adjust the learning sequence of training samples and get accurate decision boundaries. Recently, SPL has been successfully applied in the field of remote sensing. Shang et al. [31] applied SPL to detect changed regions between two SAR images and effectively reduced the performance degradation caused by noise. GSPL [32] was proposed to deal with the difficulty of collecting reliable training samples without ground labels and achieved satisfactory results in CD.

In this article, a robust self-paced incremental leaning (RSPIL) method is proposed to solve the class increment, domain increment, and interdomain class disappearance problems in MTRS image classification. First, the CD technique is employed to generate labels for target data, which results in a combined training set. To cope with the interdomain class disappearances, a novel entropy-inhibit loss function aims to reduce the likelihood of target domain samples being classified into disappearing categories. Distillation loss is able to avoid CF by fitting the outputs of the model on the target domain to the outputs on the source domain. Then the resulting loss function containing classification loss, distillation loss, and entropy-inhibit loss is combined with SPL to learn the combined training set from easy to hard. The majorize-minimization (MM) algorithm is employed to solve the resulting loss function. In RSPIL, the sample weights in SPL are incorporated into the nearest-average-eigenvector classifier and exemplar set management strategy to alleviate CF. Experimental results demonstrate the effectiveness of RSPIL on 24 datasets from four regions.

The organization of this article is organized as follows. First, Section II provides a concise overview of the change-detection-driven label transfer method, the CIL method, and SPL. Detailed exposition of the robust self-paced incremental learning (RSPIL) method is presented in Section III. Section IV is dedicated to demonstrating the experiments and results. Finally, Section V concludes the article with a summary of findings and outlines avenues for future research.

II. Background

In this section, change-detection-driven label transfer methods in MTRS image classification are first introduced in detail. In addition, the advantages and application scenarios of incremental learning in the field of remote sensing are discussed. Finally, the SPL regimes in remote sensing image classification are briefly introduced.

A. Change-Detection-Driven Label Transfer

CD in MTRS image classification is used to identify and analyze alterations or differences between two MTRS images and can be considered as a special binary classification task.

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Specifically, by jointly analyzing the source and target domain images, the CD is able to identify the changed and unchanged areas, where these areas correspond to pixel pairs at the same geographical location in the source and target domains. Consequently, the label information of pixel pairs in the unchanged areas can be shared. According to [16], the CD can be summarized into feature-level fusion strategy [33], [34] and decision-level fusion strategy [21]. Unsupervised CD methods mainly use the former strategy. In particular, the change indices are obtained from the MTRS images at the feature level. Then the indices can be analyzed by unsupervised methods [21]. Algorithms for decision-level fusion are divided into three main areas: postclassification comparison (PCC), Supervised direct multidate classification (DMC), and compound classification (CC). DMC and CC methods can achieve better results, but there are requirements for the construction of the dataset [35]. PCC directly analyzes the classification maps obtained by the classifier on the two images independently, and the performance of PCC is significantly impacted by the classification results.

To address the issue of interdomain category increment and disappearance in MTRS images, the Jeffreys–Matusita (JM) statistical distance is employed. As an effective quantification of the differences between categories, JM is able to detect instances where categories have either emerged or vanished in MTRS images. Once the categories of the source and target domains are determined, active learning [20] can be employed to assign labels to the most informative samples and effectively update the label information for the new categories in the target domain.

B. Class-Incremental Learning in Remote Sensing Images

Incremental learning [36], which enables models to integrate new knowledge while retaining existing information, can be categorized into Task, Domain, and CIL. In CIL, the main challenges include mitigating knowledge loss across categories, addressing data volume imbalance, and adapting to increasing categories. To overcome CF, strategies like data replay, parameter consolidation, and regularization are employed. Notably, mixed approaches often yield superior results, as demonstrated in [37], where a blend of current and previous training data enhances classifier training and feature representation. In addition, advanced techniques such as Knowledge Distillation, particularly the innovative weighted importance-based strategy using synthetic data replay, have proven effective in preventing knowledge erosion [38].

In the field of remote sensing image processing facing CIL challenges, the above methods have demonstrated reliable results. Rong et al. [39] leveraged historical information when learning new classes, making the current model to focus on relevant areas of previously learned categories. However, the aforementioned methods require that samples and labels for new classes are available and sufficient. In the field of remote sensing image classification, it is common to encounter scenarios of few-shot incremental learning, where the samples for new classes are insufficient. To address this issue, Wang et al. [40] enhance the feature extractor's discriminative ability by extracting features at multiple scales, effectively handling feature fusion for new class samples, and integrating classic feature pyramid networks to improve the handling of multiscale information. When both base class samples and new samples are scarce, Tai et al. [41] proposed a mining-distillation-prototype learning paradigm. Fortunately, in MTRS image classification, the labels and samples for new classes can be obtained through active learning [20], which allows RSPIL to address the class incremental problem through a supervised CIL paradigm.

C. Self-Paced Learning

Bengio et al. [42] proposed the curriculum learning (CL) method in 2004. In CL, data are gradually added to the training set of the model from easy to hard. Inspired by CL, Kumar et al. [30] established SPL as a mathematical expression with a theoretical foundation. The objective function of SPL can be defined as follows:

$$\min_{\boldsymbol{\theta}, \boldsymbol{v}\in[0,1]^n} E(\boldsymbol{\theta}, \boldsymbol{v}, \lambda) = \sum_{i=1}^{n} v_i L(y_i, J(\boldsymbol{x_i}, \boldsymbol{\theta})) + f(\boldsymbol{v}; \lambda) \quad (1)$$

where $\boldsymbol{v_i}$ is the SPL weight, which usually represents whether the samples are easy or not. $\boldsymbol{\theta}$ is the model parameters, and $f(\boldsymbol{v}; \lambda)$ is the self-paced function and is defined in [43]. $L(y_i, J(\boldsymbol{x_i}, \boldsymbol{\theta}))$ represents the loss function that calculates the loss between the true label y_i and the predicted label $J(\boldsymbol{x_i}, \boldsymbol{\theta})$, and λ is the pace parameter.

III. Methodology

In this section, the proposed RSPIL for MTRS image classification is introduced in detail. First, the challenges encountered in MTRS image classification as well as the proposed framework are described in Section III-A. Next, RSPIL and the associated optimization processes are introduced for MTRS image classification. Finally, the classification and exemplar set management method used in RSPIL are introduced. The notations used in this article are listed in Table I.

A. Framework of MTRS Image Classification

As depicted in Fig. 1, the proposed framework is designed to leverage the label information from the source domain to classify images in the target domain, while still maintaining satisfactory classification performance in the source domain. In this article, two images I_{src} and I_{tar} are selected from a sequence I to represent the source and target domains, respectively. It is assumed that there are a total of $t + n$ classes across all images, with n new classes introduced in the target image, where $n \geq 0$. Note that the methodology for identifying categories that have disappeared from the source domain, along with acquiring labels for new categories in the target domain, is elaborated in Section II-A. In the scenario of incremental learning, source domain data cannot be fully utilized for training in the target domain.

Overall, given source domain image I_{src}, target domain image I_{tar} and the label map of the source domain image

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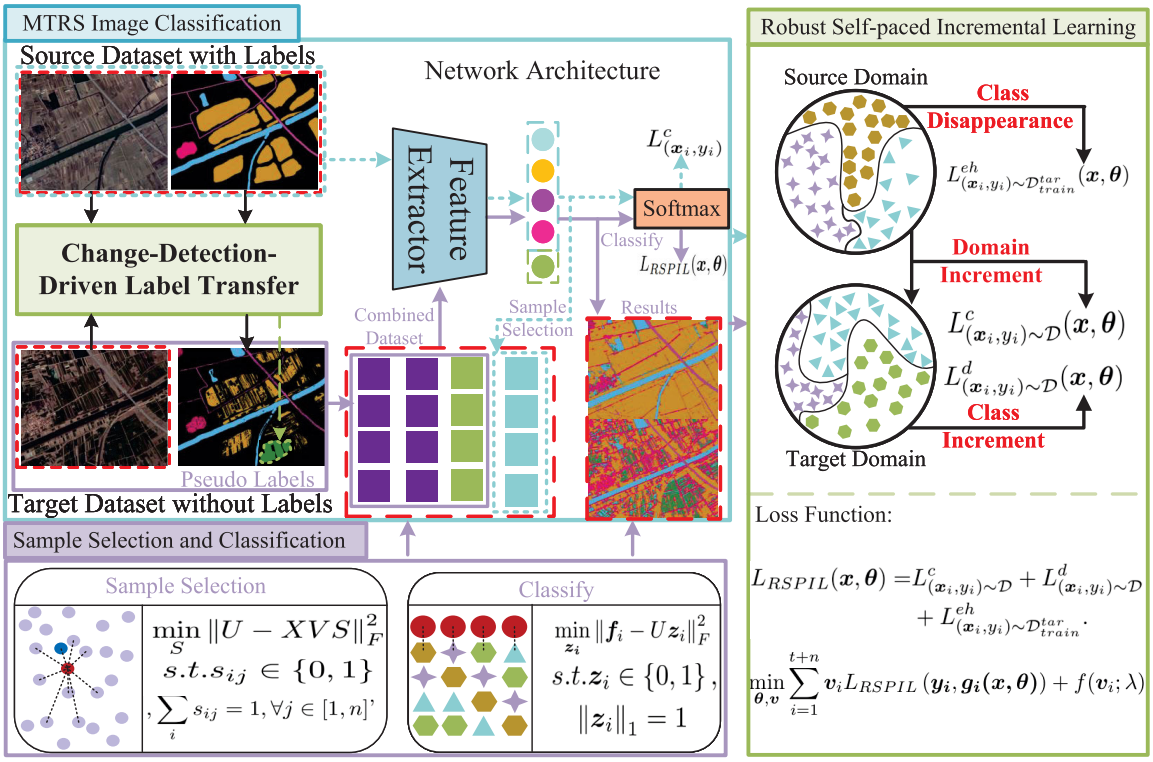


Fig. 1. Framework of RSPIL for MTRS image classification. Initially, the model is trained to convergence using source domain data, represented by blue dashed lines. Subsequently, a subset of samples is selected from the source domain data based on their logits, and these samples are integrated into the target domain training set, constructed via label transfer (depicted by purple solid lines), for joint training of the model. Finally, by employing the proposed classification method, the problem of CF can be alleviated.

TABLE I
NOTATIONS USED IN THIS ARTICLE

Notations	Descriptions
$I_{src}, I_{tar}, L_{src}, L_{tar}$	Images and corresponding classification maps of the source and target domain.
$\mathcal{X}_i^{src}, \mathcal{X}_i^{tar}, \mathcal{Y}_i^{src}, \mathcal{Y}_i^{tar}$	Source domain samples, target domain samples, source domain labels, target domain labels.
$\mathcal{D}_{train}^{src}, \mathcal{D}_{train}^{tar}, \mathcal{D}_{exem}, \mathcal{D}$	Training sets of source and target domains, $\mathcal{D}_{exem}$ is the exemplar set, $\mathcal{D}$ represents the combined training set.
$\boldsymbol{f}(\boldsymbol{x},\boldsymbol{\theta}), \boldsymbol{g}(\boldsymbol{x},\boldsymbol{\theta})$	The logits and softmax output of the model.
U, X, V, S	U represents the matrix form of the averaged eigenvector, X denotes the matrix form of the logits for all samples, V is the matrix form of sample weights, and S represents the judgment matrix.

Y_{src}, the source domain training set $\mathcal{D}_{\text{train}}^{\text{src}} = \{(\mathcal{X}_i^{\text{src}}, \mathcal{Y}_i^{\text{src}}) | i = 0, \ldots, t\}$ can be constructed using existing methods and utilized for model training. CD method introduced in Section II-A is capable of identifying unchanged areas between the source and target domains, as well as label information for new categories in the target domain, thereby enabling effective transfer of label information from the source domain to the target domain, facilitating the construction of the target domain training set $\mathcal{D}_{\text{train}}^{\text{tar}} = \{(\mathcal{X}_i^{\text{tar}}, \mathcal{Y}_i^{\text{tar}}) | i = 0, \ldots, t+n\}$. Since source domain data cannot be fully utilized, RSPIL aims to employ part of the source domain data along with the target domain data for further training, in order to simultaneously obtain classification maps for both the source and target domains.

Fig. 1 illustrates the framework of RSPIL for MTRS image classification. To prevent CF, RSPIL decouples the classifier and feature extractor and selects representative samples from the source domain data. These samples are used along with target domain samples for model training. This process is detailed in Section III-D. During the training process, RSPIL integrates classification loss, distillation loss, entropy-inhibit loss, and SPL strategy to update the model parameters, aiming to optimize the decision boundaries for each category. Note that the model trained on the source domain is referred to as the teacher model, while the one trained on the target domain is defined as the student model. The primary difference between the teacher and student models lies in their feature representation layer (the model's output layer) to accommodate potential category variations in the target domain. Since RSPIL utilizes the network for representation learning rather than actual classification steps, any existing network architecture can be applied within the proposed framework. If new categories appear in the target domain image, as illustrated in Fig. 1, new neurons corresponding to these categories will be added to the feature representation layer of the network architecture.

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Algorithm 1 MTRS Image Classification Framework

Input: MTRS images $\boldsymbol{I} = \{I_1, I_2, I_3, \cdots\}$; The source image $I_{src} \in \boldsymbol{I}$ with the labels Y_{src}; The target image $I_{tar} \in \boldsymbol{I}$ and $I_{tar} \neq I_{src}$; Hyperparameters λ_1, λ_2, K.

Output: Classification maps.

Step 1: Classification on the Source Image
Initialize SPL weights $\boldsymbol{v} = 1$.
Construct the source domain training set $\mathcal{D}_{train}^{src}$ with I_{src} and Y_{src}.
A classification model with model parameters $\boldsymbol{\theta}$ is trained based on $\mathcal{D}_{train}^{src}$ and obtain the classification map L_{src} of the source image I_{src}.
for $i = 1, \cdots, t$ **do**
    Obtain the average eigenvector $\boldsymbol{u_i}$ by Eq. (22) with $\mathcal{D}_{train}^{src}$.
end for
Obtain average eigenvectors $U = (\boldsymbol{u_1}, \ldots, \boldsymbol{u_t})$.
Obtain the exemplar set $\mathcal{D}_{exem}$ by **Algorithm 3**:

$$\mathcal{D}_{exem} = \text{ExemplarSetManagment}\left(\mathcal{D}_{train}^{src}, \boldsymbol{v}, \boldsymbol{\theta}, K\right) \tag{2}$$

Step 2: Label Transfer based on Change Detection
Identify the unchanged regions between I_{src} and I_{tar}.
Acquire the labels Y_{tar} by transferring labels from source to target in the unchanged regions.
Construct the target domain training set $\mathcal{D}_{train}^{tar}$ with I_{tar} and Y_{tar}.
Step 3: Robust Self-paced Incremental Learning
Obtain the combined training set $\mathcal{D} = \mathcal{D}_{exem} \cup \mathcal{D}_{train}^{tar}$.
Update model parameters by **Algorithm 2**:

$$\boldsymbol{\theta^*}, \boldsymbol{v} = \text{UpdateModelParameters}(\mathcal{D}, \boldsymbol{\theta}) \tag{3}$$

for $i = 1, \cdots, t + n$ **do**
    Update the average eigenvector $\boldsymbol{u_i}$ by Eq. (22).
end for
Update $U = (\boldsymbol{u_1}, \ldots, \boldsymbol{u_i}, \ldots, \boldsymbol{u_{t+n}})$ and acquire the classification map L_{tar} of I_{tar} by Eq. (23).
Update the exemplar set $\mathcal{D}_{exem}$ by **Algorithm 3**:

$$\mathcal{D}_{exem} = \text{ExemplarSetManagment}\left(\mathcal{D}, \boldsymbol{v}, \boldsymbol{\theta^*}, K\right) \tag{4}$$

The process of the proposed MTRS image classification framework is exhibited in Algorithm 2. First, the deep learning method is employed to train the model on the source domain until convergence. After this process, the average eigenvector U for each category within the source domain and the exemplar set D_{exem} can be calculated. Next, CD is applied to identify unchanged areas between the source and target domain images, and label information from these unchanged areas is transferred from the source to the target domain. Specifically, RSPIL employs CVA [44] to obtain the normalized difference map between remote sensing images from two different time phases. To obtain high-confidence labels, according to [6], the values of the normalized difference map are sorted in descending order, and the top 20% values are selected to determine the unchanged pixels. For more details on the change detection process, please refer to Section I of the supplementary materials. Label information for new categories appearing in the target domain can be acquired through [20]. A target domain training set can be constructed through these steps. Finally, the final training set $\mathcal{D}$ is formed by combining the exemplar set with the target domain training set. Utilizing this dataset, RSPIL can further update the model parameters, the average eigenvector U, and the exemplar set D_{exem}, thereby obtaining classification maps for both the source and target domains.

B. Robust Self-Paced Incremental Learning

In this section, RSPIL is proposed to deal with domain increment, class increment, and interdomain class disappearance in MTRS image classification. It should be noted that the MTRS image classification task can be considered as dense prediction task. Specifically, each pixel and its neighboring pixel in the source and target domain images are treated as individual samples. Consequently, within a single image, there can be significant variations in the number of pixels belonging to different categories, which may lead to class-imbalance issues. To address this problem, RSPIL employs the weighted cross-entropy loss function as the classification loss. The weighted multiclass cross-entropy loss can be expressed as

$$L^{c}_{(\boldsymbol{x}_i, y_i)\sim\mathcal{D}}(\boldsymbol{x}, \boldsymbol{\theta}) = -\sum_{i=1}^{t+n} \alpha_i \boldsymbol{y_i} \log \boldsymbol{g_i}(\boldsymbol{x}, \boldsymbol{\theta}) \tag{5}$$

where $\boldsymbol{y_i}$ represents the one-hot encoding form of the label, $\mathcal{D}$ represents the combined training set, and α_i is the class weight and defined as

$$\alpha_i = 1 - \frac{C_i}{\sum_{i=1}^{t+n} C_i} \tag{6}$$

where C_i is the number of pixels with the ith class. The smaller the C_i, the larger the α_i. The weight α_i makes the model pay more attention to the classes with fewer samples. $\boldsymbol{g_i}(\boldsymbol{x}, \boldsymbol{\theta})$ represents the softmax output of the samples through the model, which is shown as

$$\boldsymbol{g_i}(\boldsymbol{x}, \boldsymbol{\theta}) = \frac{\exp(\boldsymbol{f_i}(\boldsymbol{x}, \boldsymbol{\theta}))/\mathcal{T})}{\sum_{j=1}^{t+n} \exp\left(\boldsymbol{f_j}(\boldsymbol{x_i}, \boldsymbol{\theta})/\mathcal{T}\right)} \tag{7}$$

where $\mathcal{T}$ represents the distill parameter, $\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{\theta})$ represents the logits of the model. The output is the normal softmax function when $\mathcal{T} = 1$. When $\mathcal{T} > 1$, classes with lower probability can also have an effect on the loss function, making the representation contain more information about negative labels. The distillation loss is

$$L^{d}_{(\boldsymbol{x}_i, y_i)\sim\mathcal{D}}(\boldsymbol{x}, \boldsymbol{\theta}) = -\sum_{i=1}^{t+n} \boldsymbol{q_i} \log \boldsymbol{g_i}(\boldsymbol{x}, \boldsymbol{\theta}) \tag{8}$$

where $\boldsymbol{q_i}$ is the softmax output of the teacher model and $\boldsymbol{g_i}(\boldsymbol{x}, \boldsymbol{\theta})$ represents the softmax output of the student model in the situation of $\mathcal{T} = t$. Note that the teacher model is the model trained on the source domain, while the student model is the model trained on the target domain. The classification

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loss and distillation loss ensure the classification performance and minimal forgetting and can be expressed as

$$l(\boldsymbol{x},\boldsymbol{\theta}) = \sum_{(\boldsymbol{x}_i,y_i)\in\mathcal{D}} \left[L^{c}_{(\boldsymbol{x}_i,y_i)\sim\mathcal{D}} + L^{d}_{(\boldsymbol{x}_i,y_i)\sim\mathcal{D}}\right]. \tag{9}$$

The classification loss aims to minimize the discrepancy between the model predictions and the corresponding labels in the target domain, thereby facilitating the acquisition of target domain knowledge. On the other hand, the distillation loss serves to mitigate the effects of CF. Specifically, as illustrated in (8), the distillation loss employs the model's predictions on the source domain data as soft labels to aid the training process in the target domain. This approach compels the model to adopt similar feature representations and mapping techniques to those learned during its training on the source domain. Consequently, by incorporating information from the source domain, the distillation loss can effectively alleviate CF.

Considering the issue of class disappearance across domains, where some categories exist in the source domain but are absent in the target domain, the model needs to reduce the likelihood of misclassifying target domain samples into these nonexistent categories. This is because classifying samples into nonexistent categories constitutes a misclassification. To address this issue, the model needs to restrain its ability to classify target domain samples into these categories. Consequently, RSPIL introduces entropy-inhibit loss to tackle this problem. The mathematical expression of entropy-inhibit loss is

$$L^{\text{eh}}_{(\boldsymbol{x}_i,y_i)\sim\mathcal{D}^{\text{tar}}_{\text{train}}}(\boldsymbol{x},\boldsymbol{\theta}) = \sum_{d}(1-\log(1-\boldsymbol{g}_{\boldsymbol{d}}(\boldsymbol{x},\boldsymbol{\theta}))) \tag{10}$$

where $\boldsymbol{g}_{\boldsymbol{d}}(\boldsymbol{x},\boldsymbol{\theta}))$ denotes the softmax outputs for the disappearing categories based on the training samples from the target domain. These outputs can be interpreted as the model's predictive probabilities for each category. Ideally, if a category has disappeared in the target domain, the model's predictive probability for this category on the target domain is expected to trend toward 0. Therefore, if the model predicts a high probability, close to 1, the loss increases, compelling the model to push these probabilities toward 0. The final loss function can be expressed as

$$L_{\text{RSPIL}}(\boldsymbol{x},\boldsymbol{\theta}) = L^{c}_{(\boldsymbol{x}_i,y_i)\sim\mathcal{D}} + L^{d}_{(\boldsymbol{x}_i,y_i)\sim\mathcal{D}} + L^{\text{eh}}_{(\boldsymbol{x}_i,y_i)\sim\mathcal{D}^{\text{tar}}_{\text{train}}}. \tag{11}$$

Domain increment leads to variations in the feature distributions of identical classes across domains, thereby increasing the complexity of the training set and complicating the model's learning of accurate decision boundaries. Therefore, in order to get more accurate decision boundaries through sample features, RSPIL automatically selects samples from easy to hard. It is a natural idea that sample loss can be used as a criterion to evaluate the difficulty of samples.

There are a total of t classes in the source domain and n new categories in the target domain. Sample weights $\boldsymbol{v} = (\boldsymbol{v_1},\ldots,\boldsymbol{v_i},\ldots,\boldsymbol{v_t})$ in the category corresponding to the label i can be expressed as $\boldsymbol{v}_i$, where $\boldsymbol{v}_i$ is a vector where all values are constants ranging between 0 and 1, and the length of the vector is equal to the number of samples for the ith class. The reformulated objective is

$$\min_{\boldsymbol{\theta},\boldsymbol{v}} \sum_{i=1}^{t+n} \boldsymbol{v}_i L_{\text{RSPIL}}(\boldsymbol{y_i},\boldsymbol{g_i}(\boldsymbol{x},\boldsymbol{\theta})) + f(\boldsymbol{v}_i;\lambda) \tag{12}$$

where $f(\boldsymbol{v}_i;\lambda)$ is self-paced function, λ is hyperparameter to control the learning pace.

C. Optimization of RSPIL

The objective function of RSPIL is shown in (12), where there are two parameters to be optimized, the sample weight $\boldsymbol{v}$ and the model parameter $\boldsymbol{\theta}$. The MM algorithm is utilized to solve the objective function. MM transforms complex optimization problems into simple optimization problems by using the alternative iteration optimization strategy. $\boldsymbol{v}^*_i(\lambda;\mathcal{L})$ denotes the optimal weights of class i, and the integral function with respect to $\boldsymbol{v}_i(\lambda;\mathcal{L})$ can be expressed as

$$F_\lambda(\mathcal{L}) = \sum_{i=1}^{t+n}\int_0^{\mathcal{L}} \boldsymbol{v}_i(\lambda;\mathcal{L})d\mathcal{L}. \tag{13}$$

It has been proved that the tractable surrogate $Q^{(n)}_\lambda(\boldsymbol{\theta}\mid\boldsymbol{\theta}^*)$ for $F_\lambda(\mathcal{L}(\boldsymbol{\theta}))$ can be expressed as

$$Q^{(n)}_\lambda(\boldsymbol{\theta}\mid\boldsymbol{\theta}^*) = F_\lambda(\mathcal{L}(\boldsymbol{\theta}^*)) + \sum_{i=1}^{t+n}\boldsymbol{v}^*_i(\lambda;\mathcal{L}(\boldsymbol{\theta}^*))(\mathcal{L}(\boldsymbol{\theta})-\mathcal{L}(\boldsymbol{\theta}^*)). \tag{14}$$

Then, the following inequality can be easily derived:

$$\sum_{n=1}^{N} F_\lambda(\mathcal{L}(\boldsymbol{\theta})) \le \sum_{n=1}^{N} Q^{(n)}_\lambda(\boldsymbol{\theta}\mid\boldsymbol{\theta}^*). \tag{15}$$

The above process demonstrates that $Q^{(n)}_\lambda(\boldsymbol{\theta}\mid\boldsymbol{\theta}^*)$ represents a tractable alternative to $F_\lambda(\mathcal{L}(\boldsymbol{\theta}^*))$. For more information on this theorem, please refer to [45]. For convenience of the following description, $\boldsymbol{\theta}^t$ represents the model parameters after t iterations.

1) Majorization Step: The problem of obtaining $v^{m^*}_i(\lambda;\mathcal{L}(\boldsymbol{\theta}^t))$ can be transferred into finding the solution to obtain each $Q^{(n)}_\lambda(\boldsymbol{\theta}\mid\boldsymbol{\theta}^t)$

$$v^{m^*}_i(\lambda;\mathcal{L}(\boldsymbol{\theta}^t)) = \arg\min_{v^m_i\in[0,1]} \sum_{i=1}^{t+n} v^m_i\mathcal{L}(\boldsymbol{\theta}^t) + f(v^m_i;\lambda). \tag{16}$$

Several efficient self-paced functions are proposed in [43], [46] according to the above theorem. In this article, the mixture weighting scheme is selected as the self-paced function of RSPIL. When the sample loss is small, the mixture weighting scheme is able to assign a larger weight to the sample compared with other self-paced functions. The mixture weighting scheme in RSPIL is shown as follows:

$$f(v^m_i;\lambda) = -\sum_{i=1}^{t+n}\zeta\log\left(v^m_i + \frac{1}{\lambda_1}\zeta\right) \tag{17}$$

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where $\zeta = (\lambda_1\lambda_2/\lambda_1 - \lambda_2)$ and $\lambda_1 > \lambda_2 > 0$. Then, the closed-form optimal solution is given by

$$v_i^{m*} = \begin{cases} 1, & \mathcal{L} \le \lambda_2 \\ 0, & \mathcal{L} \ge \lambda_1 \\ \dfrac{\zeta}{\mathcal{L}} - \dfrac{\zeta}{\lambda_1}, & \text{otherwise} \end{cases} \tag{18}$$

where $\mathcal{L}$ is the learning objective function of RSPIL. The weight parameter v_i^{m*} is controlled by two hyperparameters λ_1, λ_2. When $\mathcal{L} < \lambda_1$, the sample is considered easy and the corresponding weight is assigned to 1. When $\mathcal{L} > \lambda_2$, the sample is considered difficult and the corresponding weight is assigned to 0. This part of the samples will not be involved in the training process. When $\lambda_1 < \mathcal{L} < \lambda_2$, the sample weight is calculated by (18).

2) Minimization Step: Sample weights are calculated through (18). After the sample weights are fixed, the model parameters should be optimized. The objective function can be expressed as

$$\boldsymbol{\theta}^{t+1} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{t+n} \boldsymbol{v}_i \mathcal{L}(\boldsymbol{\theta}^t). \tag{19}$$

The optimization of (19) can be transferred into optimizing the original loss function of RSPIL, where the only difference is the introduction of sample weights. The error function of RSPIL in the minimization step can be expressed as

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^{t+n} \boldsymbol{v}_i L(y_i, \boldsymbol{g}_i(\boldsymbol{x}, \boldsymbol{\theta})). \tag{20}$$

Consider gradient descent

$$\Delta\boldsymbol{\theta} = \eta \sum_{i=1}^{t+n} \boldsymbol{v}_i \left(\frac{\partial L^c_{(\boldsymbol{x}_i, y_i)\sim\mathcal{D}}}{\partial\boldsymbol{\theta}} + \frac{\partial L^d_{(\boldsymbol{x}_i, y_i)\sim\mathcal{D}}}{\partial\boldsymbol{\theta}} + \frac{\partial L^{\text{eh}}_{(\boldsymbol{x}_i, y_i)\sim\mathcal{D}^{\text{tar}}_{\text{train}}}}{\partial\boldsymbol{\theta}} \right) \tag{21}$$

where η is the learning rate of the model. The training process can be shown in Algorithm 2. Because of the domain increment, the training set on the target domain contains samples that follow data distributions on both the source and target domains. By integrating the loss of these samples to adjust their weights and limiting the influence of samples during model training, feature extraction in RSPIL can provide more accurate feature representations, and obtain more accurate decision boundaries.

D. Classification and Exemplar Set Management in RSPIL

It has been proposed that the coupling between the softmax classifier and the feature extractor may be the reason for CF [37]. It is a natural idea that using the classifier that decouples from the feature extractor may help alleviate CF. Nearest-average-eigenvector classifier [37], [47] selects the average eigenvector provided by the feature extraction that is closest to the sample as its belonging category. Meanwhile, the exemplar set management method in RSPIL uses a greedy strategy to obtain the sample weights of the source domain samples [48].

Algorithm 2 Update Model Parameters

Input: Training set $\mathcal{D}$; Model parameters $\boldsymbol{\theta}$.
Output: Model parameters $\boldsymbol{\theta}^*$; SPL weights $\boldsymbol{v}$.

```
for i = 1, ..., t + n do
    q_i ← f_i(x_i, θ) for all (x_i, ·) ∈ D.
end for
while Not converged do
    while Not converged do
        Update θ* by Eq. (21):
                θ* = θ − Δθ.
        Update v by Eq. (18).
    end while
    Increase λ1, λ2.
end while
```

1) Classification of RSPIL: The approximate clustering form of the classifier makes it robust to changes in feature representation. Different from the nearest-average-eigenvector classifier, after the introduction of the sample weights, the average eigenvector $U = (\boldsymbol{u}_1, \ldots, \boldsymbol{u}_i, \ldots, \boldsymbol{u}_{t+n})$ can be expressed as

$$\boldsymbol{u}_i = \frac{\boldsymbol{v}_i \boldsymbol{f}_i(\boldsymbol{x}, \boldsymbol{\theta})}{\sum \boldsymbol{v}_i}. \tag{22}$$

The objective of the classification algorithm is to find a group of $\boldsymbol{z_i}$ such that the distance between the sample feature and its class average eigenvector is the shortest, then the cluster average eigenvector corresponding to the sample is the label of the sample. For the ith sample, the classification problem can be expressed as follows:

$$\begin{aligned} &\min_{z_i} \|\boldsymbol{f}_i - U z_i\|_F^2 \\ &\text{s.t. } z_i \in \{0, 1\}, \|z_i\|_1 = 1 \end{aligned} \tag{23}$$

where $\boldsymbol{f}_i = f(\boldsymbol{x}_i, \boldsymbol{\theta}^*)$ represents the logits of samples need to be classified. Equation (23) can be solved with the exhaustive search method and the solution is the column vector with the smallest value. In particular, z_i is a binary vector and the element 1 in its column decides the category of the sample.

2) Exemplar Set Management of RSPIL: The exemplar set management method in RSPIL uses a greedy method to select source domain samples and retain their weights. The rest of the sample weights are set to 0 to ensure appropriate source domain information is used. Specifically, exemplar set management in RSPIL retains the weights of the closest samples by computing the distance between the samples and the corresponding cluster averaged eigenvectors [48].

During the construction of the exemplar set, the mathematical expression is

$$\begin{aligned} &\min_S \|U - \text{XVS}\|_F^2 \\ &\text{s.t. } s_{ij} \in \{0, 1\}, \quad \sum_i s_{ij} = 1 \quad \forall j \in [1, n] \end{aligned} \tag{24}$$

where $s_{ij} \in S$, $S \in \mathbb{S}^{n*t}$, $X \in \mathbb{X}^{d*n}$, $U \in \mathbb{U}^{d*t}$, $V \in \mathbb{V}^{n*n}$ and S is chosen to minimize the distance between X and its corresponding cluster averaged eigenvector U. U represents

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Algorithm 3 Exemplar Set Management

Input: The train set $\mathcal{D}$, memory size K, model parameters $\boldsymbol{\theta}$, SPL weights V^*, average eigenvector U^*.
Output: $\mathcal{D}_{\text{exem}} \in \mathcal{D}$
for Each sample **do**
  $\boldsymbol{f_i} \leftarrow \boldsymbol{f_i}(\boldsymbol{x_i}, \boldsymbol{\theta})$.
end for
Obtain $X = (\boldsymbol{f_1}, \ldots, \boldsymbol{f_i})$
for $k = 1, \ldots,$ K **do**
  Get contribution parameter S_Ω by Eq. (26) and Eq. (25).
end for
sort the $s_\Omega = \{s_{\Omega_1}, \ldots, s_{\Omega_K}\} \in S_\Omega$
$X_{\text{exem}} \leftarrow V_{\text{exem}} \leftarrow s_\Omega$
return $\mathcal{D}_{\text{exem}}$.

the matrix form of the averaged eigenvector, X denotes the matrix form of the logits for all samples, V is the matrix form of sample weights, and s represents the judgment matrix. Note that V is a diagonal matrix, and the value on the diagonal is $\boldsymbol{v}$. The solving equation can be expressed as

$$\min_{S_k} U - \left(X\text{VS}_k + \sum_{i=1}^{k-1} X\text{VS}_i \right) \left(\sum_{i=1}^{k} S_i^T \text{VS}_i \right)^{-1} \tag{25}$$

where k represents that the samples are selected after k iterations. The contribution parameter S_Ω can be expressed as

$$S_\Omega = \left(e^{\left(\sum_{i=1}^{k} S_i^T \text{VS}_i \right)^{-1}} - 1 \right) \left(e^{\frac{1}{k}} + \frac{1}{k} \right). \tag{26}$$

Equation (26) provides the contribution value for each sample, which represents the degree to which the sample fits the average feature vector of its respective category [37]. Since RSPIL selects samples through matrix operations and a greedy method, at least one sample is selected in each iteration. This results in more than K samples being chosen after K iterations, exceeding the maximum memory size K. Therefore, it is necessary to calculate the contribution parameter for each sample, sort them in descending order, and select the top K spatial positions as the exemplar set.

Algorithm 3 shows how to construct the exemplar set. First, the model converts each sample into a feature representation. Second, the contribution parameter for each sample is calculated based on (26). The top K samples with the highest contribution parameters are then selected to form the training set. It is important to note that the contribution parameter of each sample not only reflects the weight of the sample but also indicates the priority of its selection. In addition, sample weights are computed based on categories, ensuring that samples from each category are considered equitably. This promotes a balanced representation of samples across all categories in the final selected sample set.

IV. Experimental Study

In this section, the experimental settings will be introduced first. Next, experiments were conducted on 24 datasets to verify the effectiveness of the proposed RSPIL. Finally, RSPIL is compared with other state-of-arts MTRS classification methods.

A. Experimental Settings

1) Datasets: Four multitemporal remote sensing image sequences were designed to simulate domain increment, class increment, and interdomain class disappearance issues. Datasets used in this article were obtained through Google Earth, a powerful Earth observation software that provides high-quality satellite and aerial imagery, as well as the integration of satellite and map data. In addition, Google Earth provides a wealth of historical image data, making it very convenient for us to obtain MTRS images.

Figs. 2–5 show the MTRS image sequences. In general, for two images acquired at different times, the earlier image is used as the source domain. For each pair of images, one can be treated as the source domain and the other as the target domain. The model is trained on the source domain until convergence before training on the target domain. In this way, 24 independent datasets containing domain increments from the source to the target domain are conducted. Thus, all 24 datasets can be used for domain incremental learning. In addition, among the six datasets constructed for the Cangzhou image sequence, two datasets feature category increments from the source domain to the target domain, with a base class of 4 and an incremental setting of 1. For the Changchun dataset, four datasets include both category increments and category disappearances, with a base class of 4, an incremental setting of 1, and a disappearance of 1 category. The second part of the supplementary materials provides detailed information about the datasets. To ensure fairness, all experiments are conducted under the same settings. The experiments use the Adam optimizer with a learning rate of 0.0003 and weight decay of 0.0001. The batch size is set to 64 and the number of training epochs is set to 200. RSPIL is implemented by Torch and trained on a 16G NVIDIA RTX 3070ti GPU.

The settings of hyperparameters λ_1 and λ_2 are refer to [46]. Specifically, at the tth iteration, we can obtain $\mathcal{L}$ by sorting the loss in ascending order and selecting the nth loss value of $\boldsymbol{\mathcal{L}}$ as the value λ_1^t. Thus, λ_1^t can be represented as $\lambda_1^t = \mathcal{L}_n$. Similarly, λ_2^tcan be represented as $\lambda_2^t = 0.5 \times \mathcal{L}_n$. For classification, simple linear iterative clustering [49] is employed to divide the remote sensing images into multiple superpixel blocks. Patches from the center of the superpixel blocks can be viewed as samples to train the networks.

2) Evaluation Criteria: Pixel accuracy (PA) is determined by calculating the ratio of correctly classified pixels to the total number of pixels within the image, thus providing a measure for evaluating the model's performance in pixel-level classification tasks. The performance of RSPIL is further demonstrated through mean overall accuracy (MOA) [50] and backward transfer (BWT) [50]. MOA quantifies the average performance of incremental learning models across both source and target domains, indicating the equilibrium between acquiring knowledge in the target domain and preserving

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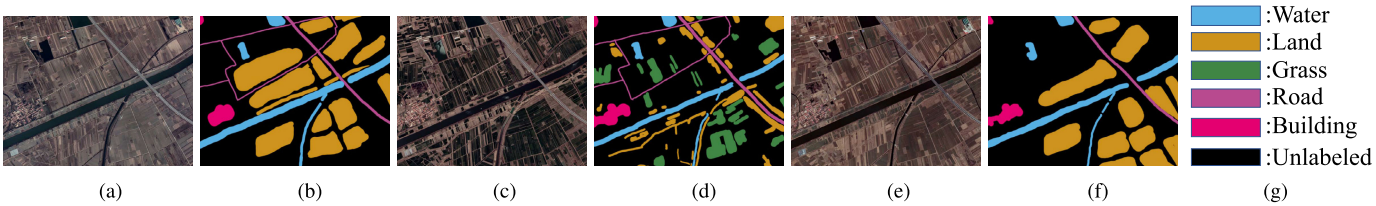


Fig. 2. Multitemporal remote sensing images of Cangzhou. t_1: April 2021, t_2: November 2021, t_3: March 2022. (a) Image (t_1). (b) Label (t_1). (c) Image (t_2). (d) Label (t_2). (e) Image (t_3). (f) Label (t_3). (g) Correspondence between the categories in the label maps (b), (d), and (f).

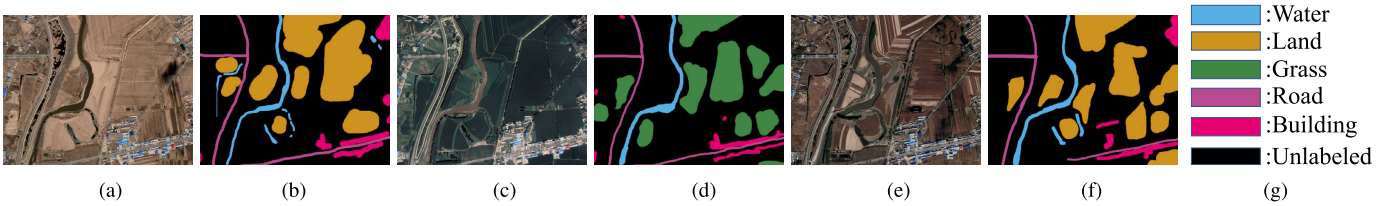


Fig. 3. Multitemporal remote sensing images of Changchun. t_1: April 2021, t_2: November 2021, t_3: March 2022. (a) Image (t_1). (b) Label (t_1). (c) Image (t_2). (d) Label (t_2). (e) Image (t_3). (f) Label (t_3). (g) Correspondence between the categories in the label maps (b), (d), and (f).

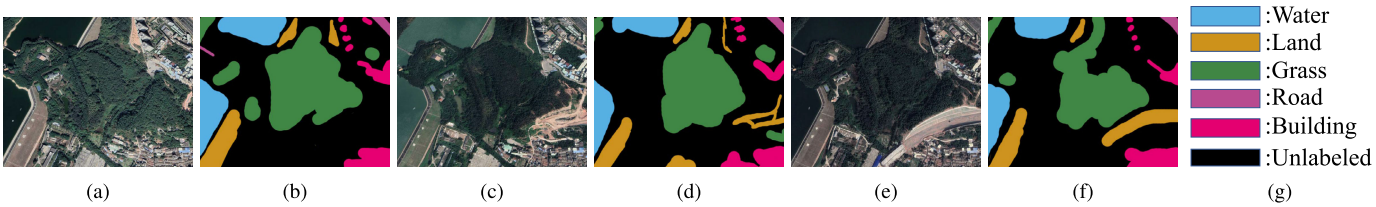


Fig. 4. Multitemporal remote sensing images of Shenzhen. t_1: December 2017, t_2: Sep. 2018, t_3: February 2022. (a) Image (t_1). (b) Label (t_1). (c) Image (t_2). (d) Label (t_2). (e) Image (t_3). (f) Label (t_3). (g) Correspondence between the categories in the label maps (b), (d), and (f).

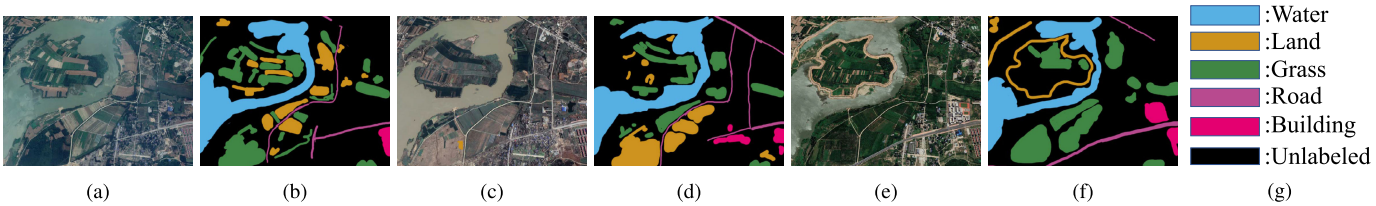


Fig. 5. Multitemporal remote sensing images of Xinyang. t_1: June 2017, t_2: March 2018, t_3: February 2022. (a) Image (t_1). (b) Label (t_1). (c) Image (t_2). (d) Label (t_2). (e) Image (t_3). (f) Label (t_3). (g) Correspondence between the categories in the label maps (b), (d), and (f).

knowledge from the source domain. The calculation formula for MOA is

$$\mathrm{MOA} = \frac{(\mathrm{PA})_s + (\mathrm{PA})_t}{2} \tag{27}$$

where $(\mathrm{PA})_s$ represents the PA result on source image and $(\mathrm{PA})_t$ represents the PA result on target image.

BWT [50] represents the forgetting of RSPIL on the source domain after learning about the target domain

$$\mathrm{BWT} = (\mathrm{PA})_s - (\mathrm{PA})_s^* \tag{28}$$

where $(\mathrm{PA})_s^*$ represents the performance of the model after training on the source domain. BWT is generally negative. The larger the BWT, the less the model forgets about knowledge from the source domain.

3) Compared Methods: Since MTRS image classification can be considered as a pixel-level classification problem, the proposed RSPIL is compared with five DA algorithms and six incremental learning methods, including four CD-driven methods: a transfer learning method CDTL [51], a consolidation parameter-based method CD-EWC [22], and two distillation-based methods: CD-LWF [52] and CD-ICARL [37], two semi-supervised DA methods: SLSA [53], E-TCA [54] and MEMO [55], FOSTER-RMM [56], and FOSTER [57]. Details of the compared algorithms are summarized as follows. Note that the label information of the target domain is obtained through the change-detection-driven label transfer method in the following methods.

1) CDTL [51], which employs CD to transfer labels from the source domain to the target domain, thereby constructing a new training set. Subsequently, this training set is utilized to update model parameters, consequently improving classification performance in both the source and target domains."
2) CD-EWC [22], which determines the importance of each model parameter to previous tasks by calculating the Fisher information matrix. A regularization term is incorporated into the loss function of the model training

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to restrict updates to parameters significant to old tasks, thus mitigating the phenomenon of CF.

3) CD-LWF [52], which integrates knowledge distillation with model fine-tuning. It trains the network using samples from the source domain and introduces a regularization term into the loss function to encourage the model to maintain the learning of the class conditional probability distribution of the source domain, thereby aiding in reducing the forgetting of old knowledge while acquiring new tasks.
4) CD-ICARL [37], which retains part of the data from the source image and incorporates distillation, mitigates CF by replaying the data from the source domain during the training process on the target domain.
5) MEMO [55], which combats CF by sharing shallow layers in neural networks that learn generalized features, while adding specialized layers for each new task to capture specific new category features, achieving an efficient balance between functionality expansion and memory usage. Moreover, MEMO minimizes memory usage by only storing necessary samples, avoiding the need to store the entire model or a large number of old task samples.
6) FOSTER-RMM [56], an extended version of FOSTER, effectively manages and maintains data memory in the incremental learning process by managing the memory rate and compression rate.
7) SLSA [53], for subspace linear semantic alignment of HSIs classification, combines linear supervised transfer learning (LSTL) [58] and SA and significantly improves the alignment performance by introducing constant terms in the learnable linear transformations.
8) FOSTER [57], which employs a strategy of feature enhancement and feature compression to handle the learning and forgetting of new and old categories. This method first improves the learning effect of new categories by enhancing features and then removes unnecessary parameters and feature dimensions through a compression process to maintain the model's memory of old categories, thus combating CF.
9) E-TCA [54], aims to select a discriminative and consistent subset of features from the source domain by learning a feature transformation matrix that maximizes the similarity between the source domain and the target domain.

B. Experiments on the Proposed RSPIL

To further investigate RSPIL, two experiments were conducted on the 24 datasets. First, the forgetting degree of CD-based continual learning methods and RSPIL were tested on 24 datasets. Statistical tests were then conducted on the experiment results to validate the effectiveness of RSPIL in mitigating CF. Finally, the effectiveness of the proposed entropy-inhibit loss function was tested. In the experiments, a concised CD algorithm which combines change vector analysis (CVA) and K-means was selected and represented as PCAKM [59], [60]. Due to the simplicity in computation and the ability to effectively identify meaningful changes, PCAKM is widely applied to CD-based MTRS image classification algorithms [60], [61], [62]. To ensure fairness, all experiments utilized PCAKM as the CD algorithm.

TABLE II
RESULTS OBTAINED BY FIVE METHODS IN TERMS OF BWT(%). THE BEST VALUE IS IN BOLD

Dataset	**CD-EWC** [22]	**CD-LWF** [52]	**CD-ICARL** [37]	**CDTL** [51]	**RSPIL**
Cangzhou-(t_1,t_2)	-76.71	-61.86	-4.62	-34.72	**-0.45**
Cangzhou-(t_1,t_3)	-2.17	-3.04	-1.00	-3.86	**-0.01**
Cangzhou-(t_2,t_1)	-13.05	-15.71	-28.45	-26.60	**-6.41**
Cangzhou-(t_2,t_3)	-27.64	-35.08	-32.91	-37.76	**-3.03**
Cangzhou-(t_3,t_1)	0.53	-1.42	0.04	-4.28	**1.82**
Cangzhou-(t_3,t_2)	0.62	-8.91	0.44	-10.94	**2.46**
Changchun-(t_1,t_2)	-79.63	-38.71	-3.64	-80.07	**-3.19**
Changchun-(t_1,t_3)	-9.84	-11.53	-6.21	-10.03	**-2.87**
Changchun-(t_2,t_1)	-82.30	-84.76	-3.95	-83.54	**2.00**
Changchun-(t_2,t_3)	-5.08	-54.88	-31.57	-8.50	**1.92**
Changchun-(t_3,t_1)	-9.43	-11.09	-12.64	-19.57	**-5.00**
Changchun-(t_3,t_2)	-24.71	-66.10	-15.21	-76.99	**-7.39**
Shenzhen-(t_1,t_2)	**-1.31**	-21.63	-12.03	-21.79	-4.52
Shenzhen-(t_1,t_3)	**-1.22**	-7.39	-7.12	-2.50	-3.19
Shenzhen-(t_2,t_1)	-5.60	-7.08	**-3.40**	-10.82	-5.39
Shenzhen-(t_2,t_3)	1.36	-0.41	-1.24	-2.07	**2.41**
Shenzhen-(t_3,t_1)	-0.35	-1.00	-1.54	-1.63	**0.38**
Shenzhen-(t_3,t_2)	0.25	-0.15	0.06	-4.29	**0.85**
Xinyang-(t_1,t_2)	-40.57	-55.43	-12.74	-56.21	**-1.94**
Xinyang-(t_1,t_3)	-14.36	-39.36	-1.88	-39.89	**-0.62**
Xinyang-(t_2,t_1)	-37.14	-78.56	-3.02	-72.88	**-2.39**
Xinyang-(t_2,t_3)	-16.44	-27.55	-3.39	-17.17	**0.25**
Xinyang-(t_3,t_1)	-19.78	-35.4	-8.36	-46.68	**-0.92**
Xinyang-(t_3,t_2)	-42.04	-33.27	-44.22	-48.43	**-0.98**

1) Experiments on Measuring the Forgetting: To validate the effectiveness of inhibiting CF of RSPIL, we compared the BWT metrics of CD-EWC, CD-LWF, CD-ICARL, and CDTL with RSPIL. Table II presents the forgetting degrees of the five methods across 24 datasets. From Table II, it can be observed that RSPIL outperforms the other four methods in terms of BWT on 21 datasets. Unfortunately, RSPIL does not achieve the best BWT metric on the Shenzhen-(t_1, t_2), Shenzhen-(t_1, t_3), and Shenzhen-(t_2, t_1) datasets, where it has a slight decrease of 3.21%, 1.97%, and 1.99%, respectively, compared to the best BWT. This is because, in Shenzhen datasets, the visual differences in the same categories between the source domain and the target domain are not significant, and the data distribution of these categories is similar.

However, for datasets involving class increment, domain increment, and class disappearance issues simultaneously, CIL methods such as CD-EWC, and CD-LWF perform poorly, particularly on the Changchun-(t_1, t_2) and Changchun-(t_2, t_1) datasets. Specifically, CD-EWC achieves a BWT of -79.63% on the Changchun-(t_1, t_2) dataset. This indicates that when land cover categories disappear in the target domain and there are significant visual differences between the source and target domain samples, by penalizing changes in important weights may lead to insufficient flexibility to adapt to new categories. Similarly, LWF exhibited a BWT of -84.76% on the Changchun-(t_2, t_1) dataset, demonstrating its difficulty in finding a feature representation suitable

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TABLE III
RESULTS OF THE RSPIL AND RSPIL-W/O-L_{RSPIL} IN THE DATASETS CONTAIN CLASS DISAPPEARANCE

Datasets	Cangzhou-(t_2,t_1)		Cangzhou-(t_2,t_3)		Changchun-(t_1,t_2)		Changchun-(t_1,t_2)		Changchun-(t_2,t_3)		Changchun-(t_3,t_2)	
Criteria	**NOP**	**MOA**(%)	**NOP**	**MOA**(%)	**NOP**	**MOA**(%)	**NOP**	**MOA**(%)	**NOP**	**MOA**(%)	**NOP**	**MOA**(%)
RSPIL	1160	90.70	1322	93.20	0	95.20	1091	92.32	4232	90.70	350	94.64
RSPIL-w/o-L_{RSPIL}	13557	88.25	1789	90.99	136	94.47	2537	92.69	6112	91.19	624	94.96

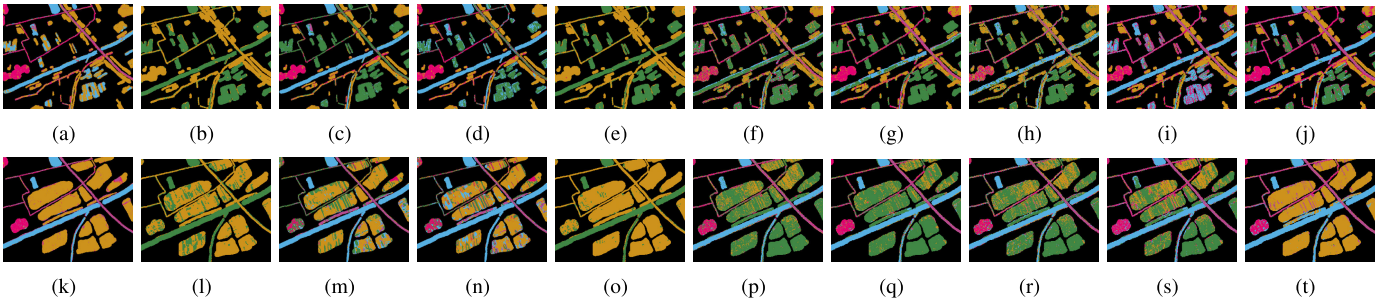


Fig. 6. Results of RSPIL compared with other methods on Cangzhou-(t_2, t_1) dataset. (a) and (k) MEMO, (b) and (l) SLSA, (c) and (m) FOSTER, (d) and (n) FOSTER-RMM, (e) and (o) E-TCA, (f) and (p) CDTL, (g) and (q) CDEWC, (h) and (r) CDLWF, (i) and (s) CDICARL, and (j) and (t) RSPIL.

for both source and target domain images when faced with substantial data distribution differences between the domains. Furthermore, CD-ICARL and RSPIL adopt exemplar set management strategies and nearest-average-eigenvector classifiers, enhancing their ability to learn new representations. Particularly on the Xinyang datasets, RSPIL significantly outperformed other methods, which further confirms that in scenarios where there are substantial data distribution differences between the source and target domains, RSPIL can effectively identify feature representations suitable for both source and target domain images, thereby mitigating the forgetting problem in the model.

2) Experiments on the Interdomain Class Disappearance: To validate the effectiveness of the proposed RSPIL in reducing the number of samples classified into disappearing classes in the target domain, experiments were designed on six datasets that exist interdomain class disappearance problem to compare the performances of RSPIL and RSPIL-w/o-L_{RSPIL}. The only difference between RSPIL-w/o-L_{RSPIL} and RSPIL is that RSPIL-w/o-L_{RSPIL} uses distillation loss and classification loss. Table III presents the MOA and the number of pixels classified into disappearing classes (NOP) on the target images across the six datasets. From Table III, it can be observed that RSPIL significantly reduces the NOP in the target images, despite having a slightly lower MOA compared to RSPIL-w/o-L_{RSPIL}. Specifically, in the datasets which contain class disappearance problems, RSPIL effectively reduces the number of pixels in the target images that are classified into the disappeared categories. On Cangzhou-(t_2, t_1) dataset, RSPIL reduces 12 397 pixels compared with RSPIL-w/o-L_{RSPIL}. Although the MOA of RSPIL is slightly 1.68% lower than RSPIL-w/o-L_{RSPIL} on Changchun-(t_2, t_3) dataset, RSPIL reduces 1880 pixels, which also proves the effectiveness of RSPIL to the interdomain class disappearance problem.

C. Comparison With Other Methods

To further verify the effectiveness of the proposed method, RSPIL is compared with nine classical MTRS image classification methods, including MEMO [56], FOSTER [57], SLSA [53], FOSTER-RMM [56], E-TCA [54], CDTL [51], CD-EWC [22], CD-LWF [52], and CD-ICARL [37]. Details of the compared algorithms are summarized at the end of Section IV-A.

1) Results on Cangzhou Datasets: The classification maps and the results of RSPIL and nine compared methods on Cangzhou datasets are shown in Fig. 6 and Table IV. In general, RSPIL achieved better performance compared to other methods, with similar phenomena occurring on Shenzhen, Changchun, and Xinyang datasets. From Fig. 6, it is obvious that CDTL, CD-EWC, CD-LWF, and CD-ICARL misclassify most of the pixels belonging to the category land as grass. This is because the visual differences between these two categories are not significant, making it difficult for the model to distinguish their discriminative features. Specifically, in the target domain training set, the labels of samples belonging to the category land are incorrectly identified as grass. It can also be found that SLSA and E-TCA methods all have low classification accuracy in the category of water and building in the target domain. The similar features between the category of grass in the source image and the category of water in the target image result in the misclassification.

Table IV presents the classification results of ten methods on Cangzhou, Changchun, Shenzhen, and Xinyang datasets. On Cangzhou-(t_1, t_3) and Cangzhou-(t_3, t_1) datasets, most methods get satisfactory results. This is because the two datasets only exhibit domain increment issues, and there is no significant visual difference between interroad classes. Due to the class-imbalance issue, DA methods tend to classify samples into the majority classes. On the Cangzhou-(t_1, t_2) dataset, the CD-EWC method achieves only 28.87% MOA. This is because of the error of change detection, even though in experiments we tend to select samples with higher confidence. The error in change detection still transfers some incorrect label information to the target domain, which makes it difficult for the CD-EWC method to adapt to the classification in the target domain while protecting the parameters important for

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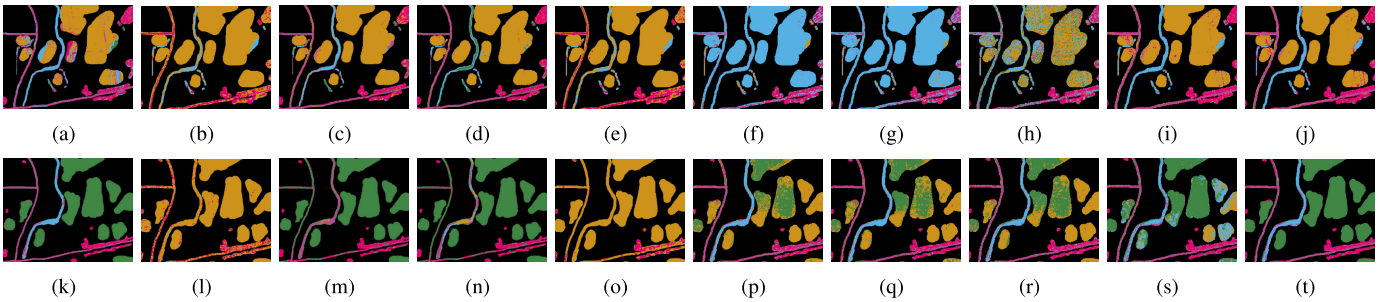


Fig. 7. Reults of RSPIL compared with other methods on Changchun-(t_1, t_2) dataset. (a) and (k) MEMO, (b) and (l) SLSA, (c) and (m) FOSTER, (d) and (n) FOSTER-RMM, (e) and (o) E-TCA, (f) and (p) CDTL, (g) and (q) CDEWC, (h) and (r) CDLWF, (i) and (s) CDICARL, and (j) and (t) RSPIL.

classification in the source domain. On the whole, it can be distinctly found that RSPIL performs the best on Cangzhou datasets. The classification maps of RSPIL match reasonably well with the ground truth, which further illustrates its superiority.

2) Results on Changchun Datasets: For Changchun datasets, Fig. 7 and Table IV, respectively, display the classification maps on Changchun-(t_1, t_2) and the results of RSPIL and compared methods mentioned above. From Fig. 7, it can be observed that DA methods fail to achieve satisfactory classification results in the target images because of the lack of annotated information. Specifically, these methods struggle to accurately classify the pixels of the grass and water categories in the target domain. Due to the significant differences between the source and target images, CD fails to provide target images with sufficient and correct annotated information. Thus, CDTL, CD-EWC, and CD-LWF methods have serious CF problems on the source domain, and poor classification performance on the target domain. In the classification maps of CDTL and CD-EWC, almost all pixels of land are wrongly identified as water. Because of the introduction of distillation loss, the CF of CD-LWF is mitigated. MEMO, CD-ICARL, and RSPIL effectively mitigate the phenomenon of CF by retaining a portion of the source domain samples through sample set management strategies. Furthermore, the retained source domain information aids these algorithms in extracting more discriminative features to adapt to the distribution changes between the source and target domain data. RSPIL gradually selects samples for training and the classification maps are the closest to the ground truth.

Table IV exhibits the MOA metrics of ten methods on six datasets located in Changchun. It can be observed that DA methods achieve only around 45% MOA in datasets that involve class increment and class disappearance issues, such as Changchun-(t_1, t_2), Changchun-(t_2, t_1), Changchun-(t_2, t_3), and Changchun-(t_3, t_2) datasets. Transfer learning and continual learning methods also fail to achieve satisfactory MOA results on these datasets. However, RSPIL consistently achieves MOA above 90%. Specifically, on the Changchun-(t_1, t_2) dataset, RSPIL is 13.99% higher than the CD-ICARL method. Similarly, on the Changchun-(t_1, t_3) dataset, RSPIL surpasses the CD-EWC method by 2.79%. In addition, FOSTER and FOSTER-RMM perform poorly on the Changchun dataset constructed from t_1, t_2 image pairs and t_2, t_3 image pairs, with MOA values ranging from 60% to 70%. This is due to the high complexity of these datasets, which encompass three issues: class incremental, domain incremental, and class disappearance problems. FOSTER has limitations in handling highly dynamic data environments. Furthermore, the feature compression strategy used in FOSTER may result in information loss when attempting to address class increment and domain increment issues. The MOA metrics further validate the superiority of RSPIL in MTRS image classification.

3) Results on Shenzhen Datasets: Table IV shows the MOA metrics of RSPIL and compared approaches are, respectively, displayed. Due to the length limitation of the article, the classification diagrams of the Shenzhen dataset are presented in the supplementary materials. It can be discovered that SLSA, and E-TCA all have no ability to distinguish the class of water and building on the target domain. Different from other datasets, there is no visual and significant difference in the image sequence, which means that CD can provide sufficient and accurate labels for the target domain. Therefore, apart from the slight CF in which CDTL misclassifies some regions covered by water, RSPIL and other continual learning methods achieve satisfactory results on the source image and target image.

With respect to the MOA metrics of Table IV, RSPIL achieves the best MOA on three datasets Shenzhen-(t_2, t_3), Shenzhen-(t_3, t_1) and Shenzhen-(t_3, t_2). However, on Shenzhen-(t_1, t_2), Shenzhen-(t_1, t_3) datasets, the MOA of RSPIL are slightly lower than CD-EWC, reaching 4.1% and 0.96%, respectively. On Shenzhen-(t_2, t_1) dataset, CD-LWF achieves the best performance of MOA. The reason is that the Shenzhen dataset is relatively simple, only exists the domain increment problem, and the influence of error propagation generated by the CD process is much smaller than that of other datasets. Overall, RSPIL does not perform best on all datasets, but still achieves satisfactory results.

4) Results on Xinyang Datasets: In the Xinyang datasets, Table IV displays the results of RSPIL compared to other methods, respectively. Due to spatial constraints, we only showcase the classification outcomes for the Xinyang-(t_1, t_2) dataset in the supplemental materials. Overall, RSPIL demonstrates superior performance over alternative approaches. For example, SLSA struggles beyond distinguishing between water and grass. CDTL, CD-EWC, and CD-LWF misclassify land pixels as grass in the target image and wrongly label water as grass in the source image. CD-ICARL reduces CF but inaccurately classifies water as land in the target domain.

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TABLE IV
RESULTS OBTAINED BY TEN METHODS IN TERMS OF MOA(%). THE BEST VALUE IS IN BOLD

Dataset	MEMO [55]	SLSA [53]	FOSTER [57]	FOSTER-RMM [56]	E-TCA [54]	CD-EWC [22]	CD-LWF [52]	CD-ICARL [37]	CDTL [51]	RSPIL
Cangzhou-(t_1,t_2)	70.06	69.78	74.65	71.70	70.26	28.87	61.62	92.34	75.56	**94.44**
Cangzhou-(t_1,t_3)	75.33	90.69	87.76	85.10	90.67	87.11	94.94	96.43	95.31	**97.87**
Cangzhou-(t_2,t_1)	79.65	62.14	69.97	66.94	68.05	58.10	59.67	55.09	54.06	**90.70**
Cangzhou-(t_2,t_3)	79.32	49.76	59.82	52.87	57.59	53.33	49.64	57.17	49.83	**93.20**
Cangzhou-(t_3,t_1)	79.22	90.90	58.57	62.54	90.90	96.28	95.47	96.17	94.29	**96.67**
Cangzhou-(t_3,t_2)	50.03	69.28	57.69	72.86	69.51	86.35	80.41	91.82	80.15	**94.22**
Changchun-(t_1,t_2)	81.26	42.59	80.83	80.87	43.56	35.12	56.16	81.21	33.37	**95.20**
Changchun-(t_1,t_3)	80.14	77.14	66.18	66.24	80.29	90.44	89.01	89.13	90.28	**93.23**
Changchun-(t_2,t_1)	83.20	43.36	63.27	65.53	48.89	15.46	17.87	66.29	16.69	**92.32**
Changchun-(t_2,t_3)	78.91	43.42	66.88	68.48	49.14	54.51	29.82	51.55	52.97	**90.70**
Changchun-(t_3,t_1)	82.26	76.30	67.78	66.29	77.50	91.17	90.34	88.87	86.27	**94.64**
Changchun-(t_3,t_2)	86.42	39.89	79.89	81.88	40.46	54.37	33.78	70.85	29.45	**93.55**
Shenzhen-(t_1,t_2)	88.78	51.43	72.98	52.17	59.95	95.08	84.62	87.71	84.70	**90.98**
Shenzhen-(t_1,t_3)	88.10	56.51	75.18	54.70	64.00	**95.13**	85.33	90.94	94.79	94.17
Shenzhen-(t_2,t_1)	80.05	61.26	74.20	53.58	64.24	91.01	91.14	**92.20**	89.09	89.86
Shenzhen-(t_2,t_3)	77.46	65.17	78.76	64.20	73.58	95.89	95.11	94.34	94.15	**96.90**
Shenzhen-(t_3,t_1)	79.52	71.00	74.01	50.67	69.62	94.49	94.33	93.26	94.03	**94.98**
Shenzhen-(t_3,t_2)	89.63	62.38	74.39	44.72	63.76	94.97	95.28	94.75	93.54	**95.56**
Xinyang-(t_1,t_2)	51.94	72.89	51.94	49.27	70.29	65.71	58.77	63.31	58.42	**92.84**
Xinyang-(t_1,t_3)	52.56	78.73	52.56	50.25	83.99	86.29	73.33	88.90	73.31	**95.18**
Xinyang-(t_2,t_1)	51.93	72.65	51.93	44.23	80.12	70.74	50.46	73.33	53.66	**92.94**
Xinyang-(t_2,t_3)	50.67	73.82	50.67	49.43	79.30	85.22	77.98	89.59	83.01	**94.76**
Xinyang-(t_3,t_1)	51.47	79.02	51.47	47.06	81.10	80.81	71.99	82.81	67.92	**95.62**
Xinyang-(t_3,t_2)	50.71	66.11	50.71	50.78	56.49	62.90	67.97	60.10	60.46	**93.46**

Table IV outlines the MOA metrics for RSPIL and other methods on the Xinyang datasets, highlighting issues of domain increment and error propagation. Unlike in the Shenzhen datasets, error propagation from the CD process is significantly worse in the Xinyang datasets, especially in the Xinyang-(t_1, t_2), Xinyang-(t_2, t_1), and Xinyang-(t_3, t_2) series. Consequently, incremental learning and transfer learning approaches based on CD achieve only around 60%–70% in MOA metrics. Specifically, MEMO has an MOA value of only 50% on these datasets. This is because there are significant visual differences between the source and target domains in the Xinyang dataset, resulting in considerable differences in their data distributions. This makes it challenging for MEMO to capture common cross-domain features from generalized shallow layers. In addition, due to the difficulty in obtaining sufficient and accurate label information for the target domain in change detection, MEMO struggles to extract discriminative features from deeper layers. A similar situation occurs with FOSTER and FOSTER-RMM. Due to insufficient label information for the target domain, FOSTER finds it difficult to accurately estimate the feature differences between the source and target domains with the limited data points. Furthermore, feature compression may lead to information loss, making it challenging for FOSTER to handle domain incremental issues. However, DA techniques generally surpass CD-based methods, with RSPIL notably achieving over 90% MOA across all examined datasets, making its classification maps closest to the actual ground truth.

V. CONCLUSION

In this article, we propose an RSPIL method to address the challenges of domain increment, class increment, and inter-domain class disappearance in MTRS image classification. First, we introduce weighted classification loss and distillation loss to tackle the issues of class increment and CF, respectively. To address interdomain class disappearance, we propose an entropy-inhibit loss. Subsequently, we develop the corresponding solvers utilizing the MM algorithm, which iteratively updates the model parameters and sample weights. Specifically, we integrate sample weights into the proposed nearest-average-eigenvector classifier and exemplar set management strategy to further mitigate CF. Experimental results on 24 datasets from four real-world locations demonstrate the outstanding performance of our methods. There are two significant aspects to consider in future research: 1) prior knowledge in experience-replay-based methods will no longer focus only on preserving representative samples, but on the characteristics of the samples and 2) other incremental learning methods will be considered for MTRS image classification problem to alleviate the CF phenomenon.

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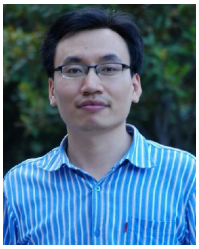
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